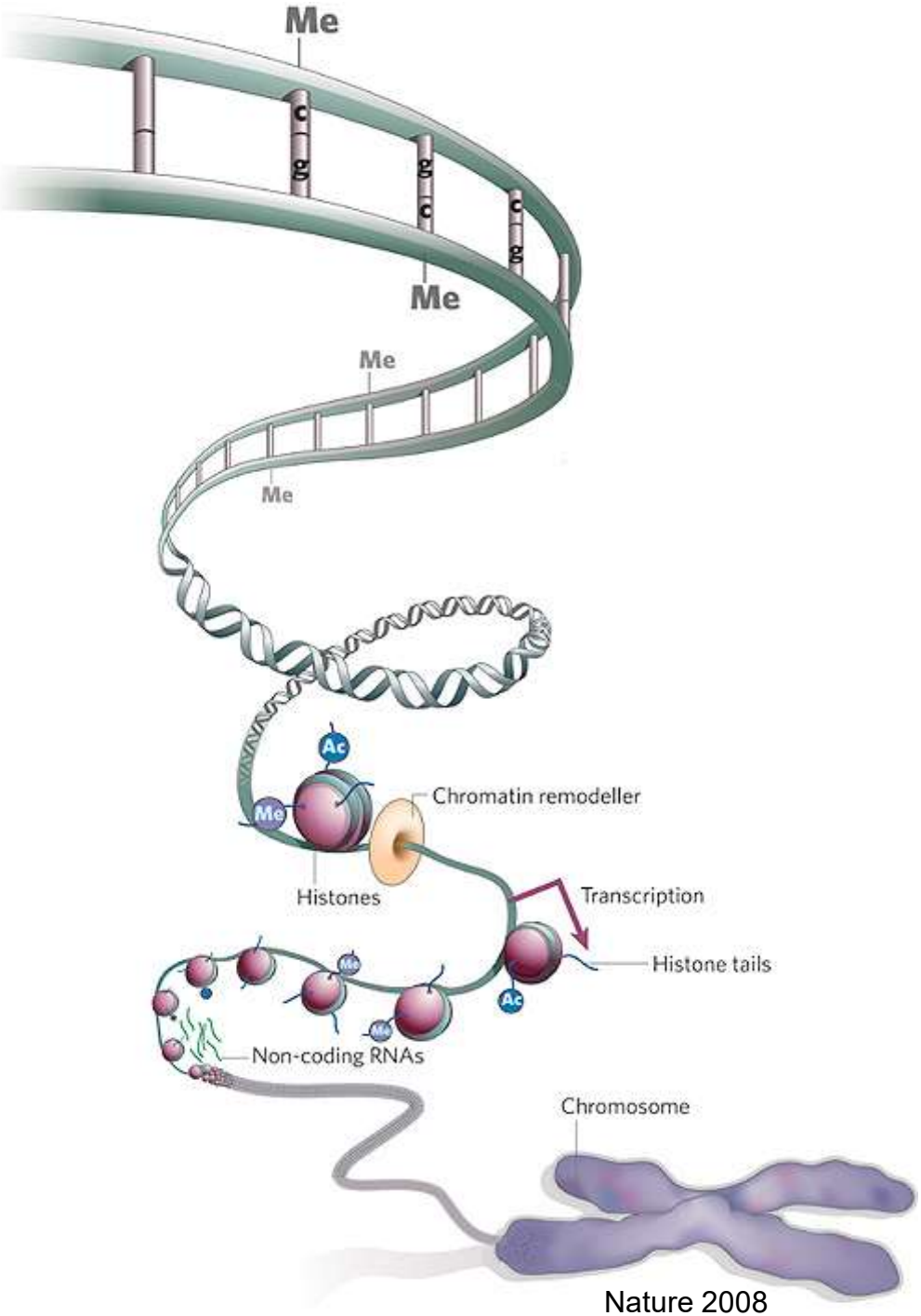
- Sequencing bias & errors
- Speed & RAM
- Data storage & visualization

Specific challenges:
(4) epigenomics

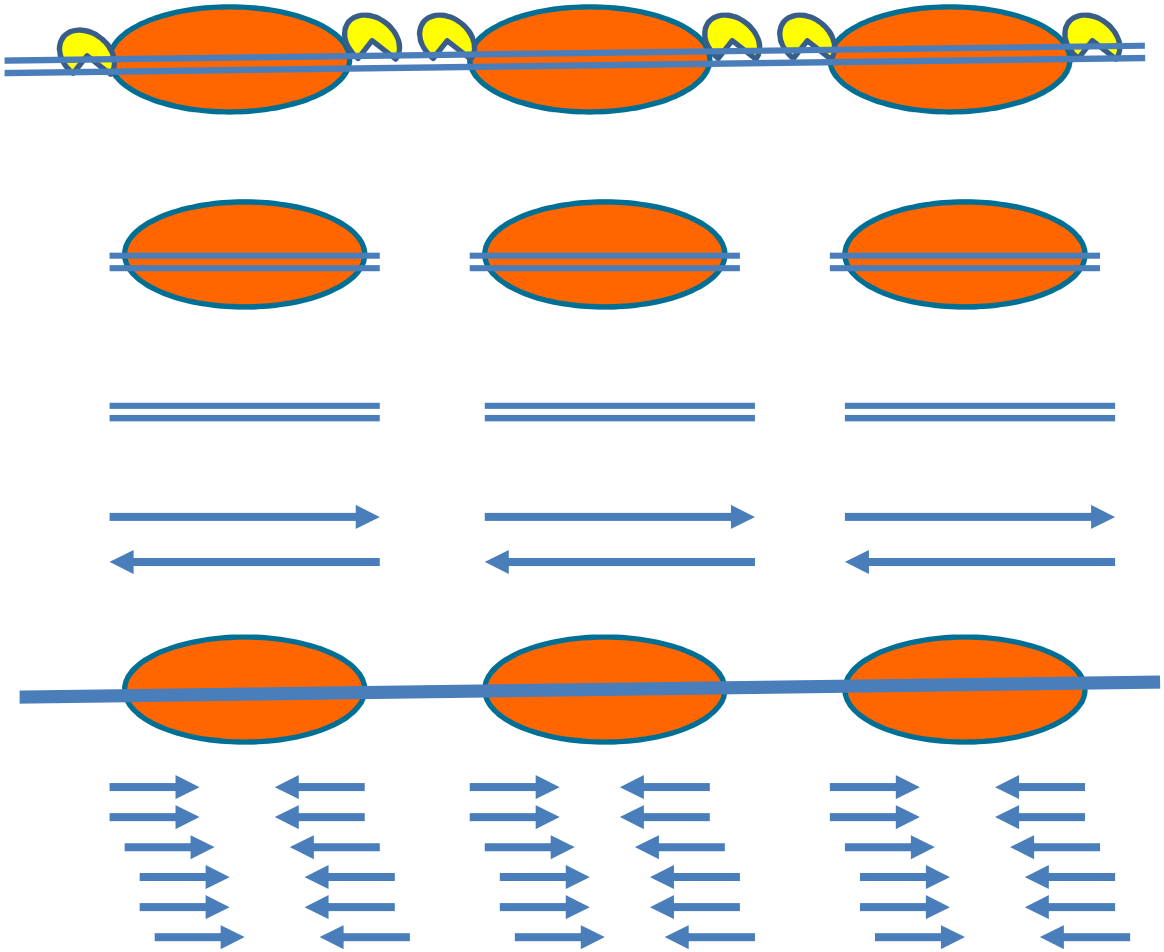
Epigenetics refers to the study of changes in the regulation of gene activity and expression that are not dependent on gene DNA sequence.

- **Protein-DNA interactions [ChIPSeq]**
  - Nucleosome positioning
  - Histone modification
  - Transcription factor interactions
- **Methylation [MethylSeq]**
- **Impact of NGS**
  - Whole genome profiling
  - Resolution
- **Computational challenges**
  - Systematic bias
  - Unambiguous mapping
  - Robust event calling



Nature 2008

ChIPSeq

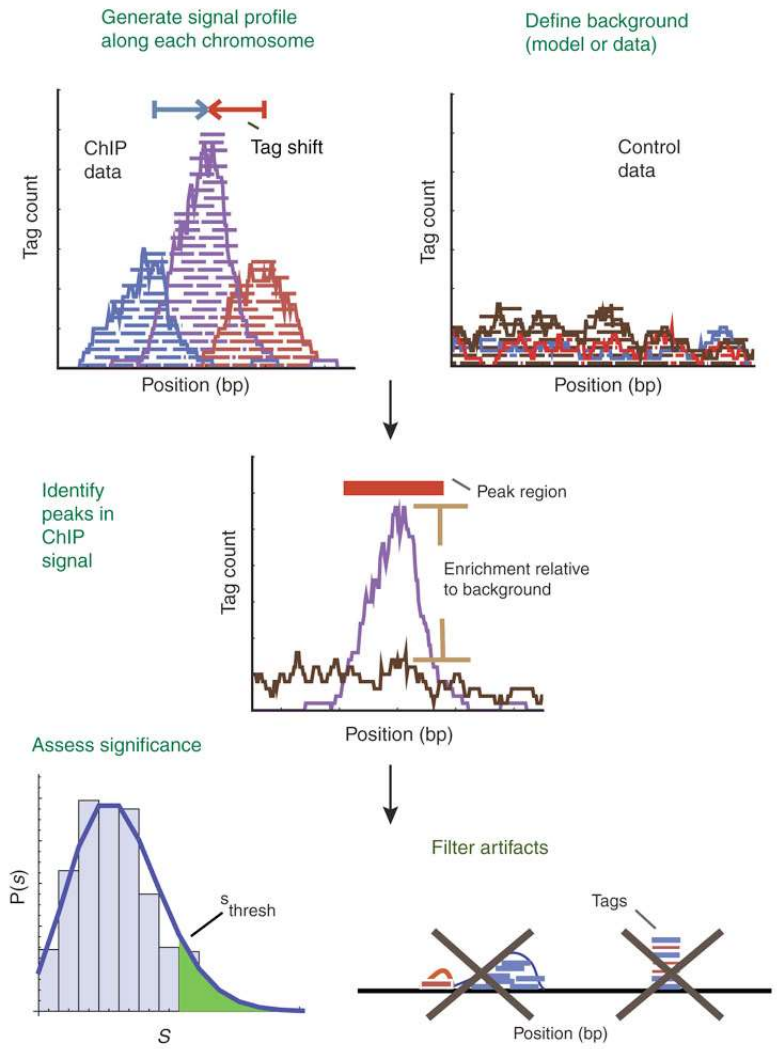


MNase
Linker Digest

Remove
Nucleosomes

Sequence &
Align

CHIP-Seq methods



CisGenome

ERANGE

FindPeaks

F-Seq

GLITR

MACS

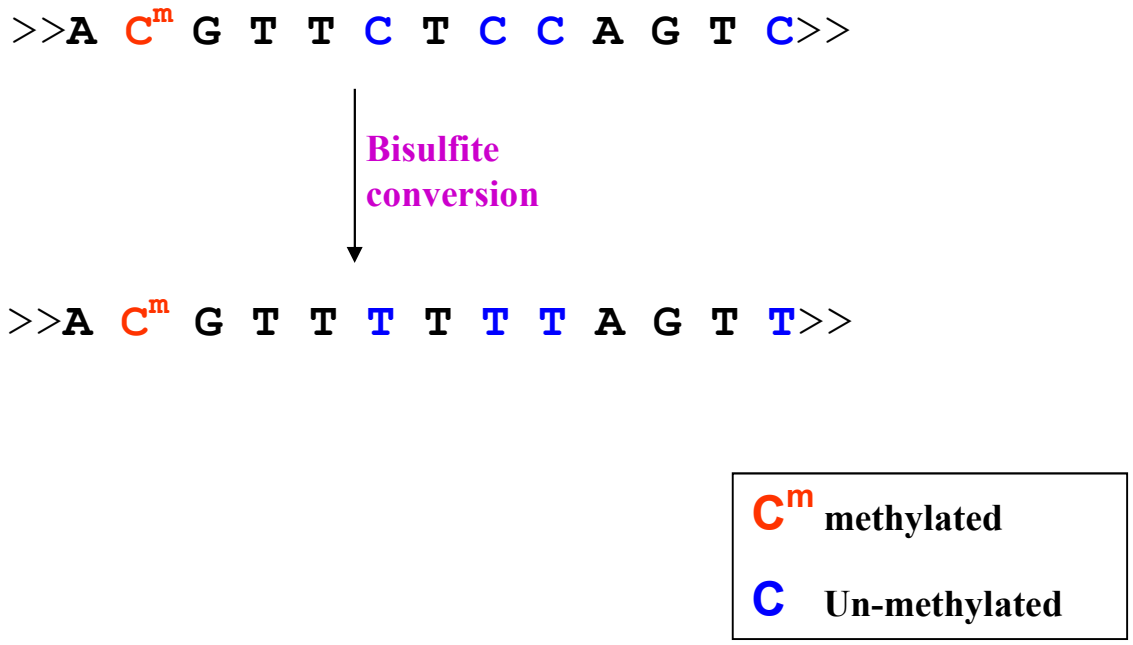
PeakSeq

QuEST

Pepke et al. Nature Methods 2009

Problems of mapping BS-seq reads

- Reduced sequence complexity



错误原因：异质性样本（批次效应）



Comparison of the transcriptional landscapes between human and mouse tissues

Shin Lin[a,b,1], Yiing Lin[c,1], Joseph R. Nery[d], Mark A. Urich[d], Alessandra Breschi[e,f], Carrie A. Davis[g], Alexander Dobin[g], Christopher Zaleski[g], Michael A. Beer[h], William C. Chapman[c], Thomas R. Gingeras[g,i], Joseph R. Ecker[d,j,2], and Michael P. Snyder[a,2]

案例分析



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Archive > Volume 521 > Issue 7553 > Research Highlights: Social Selection > Article

NATURE | RESEARCH HIGHLIGHTS: SOCIAL SELECTION

Potential flaws in genomics paper scrutinized on Twitter

Reanalysis of a study that compared gene expression in mice and humans tests social media as a forum for discussing research results.

RESEARCH ARTICLE

A reanalysis of mouse ENCODE comparative gene expression data [version 1; referees: 3 approved, 1 approved with reservations]

Yoav Gilad, Orna Mizrahi-Man

Department of Human Genetics, University of Chicago, Chicago, IL, 60637, USA

错误原因：噪音污染（附生和共生菌）



Evidence for extensive horizontal gene transfer from the draft genome of a tardigrade

Thomas C. Boothby[a,1], Jennifer R. Tenlen[a,2], Frank W. Smith[a], Jeremy R. Wang[a,b], Kiera A. Patanella[a], Erin Osborne Nishimura[a], Sophia C. Tintori[a], Qing Li[c], Corbin D. Jones[a], Mark Yandell[c], David N. Messina[d], Jarret Glasscock[d], and Bob Goldstein[a]

[a]Department of Biology, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599; [b]Department of Genetics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599; [c]Eccles Institute of Human Genetics, University of Utah, Salt Lake City, UT 84112; and [d]Cofactor Genomics, St. Louis, MO 63110

Edited by W. Ford Doolittle, Dalhousie University, Halifax, Canada, and approved September 28, 2015 (received for review May 28, 2015)



No evidence for extensive horizontal gene transfer in the genome of the tardigrade *Hypsibius dujardini*

Georgios Koutsovoulos[a], Sujai Kumar[a], Dominik R. Laetsch[a,b], Lewis Stevens[a], Jennifer Daub[a], Claire Conlon[a], Habib Maroon[a], Fran Thomas[a], Aziz A. Aboobaker[c], and Mark Blaxter[a,1]

[a]Institute of Evolutionary Biology, University of Edinburgh, Edinburgh EH9 3FL, United Kingdom; [b]The James Hutton Institute, Dundee DD2 5DA, United Kingdom; and [c]Department of Zoology, University of Oxford, Oxford OX1 3PS, United Kingdom

Edited by W. Ford Doolittle, Dalhousie University, Halifax, Canada, and approved March 1, 2016 (received for review January 8, 2016)

如何有效解读海量的高维度、多噪音、异质性

组学数据是生物信息学研究的关键问题